HW1 实验报告

Logic Synthesis

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本报告回答 ***Problem*1** 和***Problem*2** 中的问题，并介绍***Problem*3** 的算法设计思路

1 Problem 1

***Prove*：**

$$\begin{aligned} D_x(f \wedge g) &= (f \wedge g)(x=1, \boldsymbol{u}) \oplus (f \wedge g)(x=0, \boldsymbol{u}) \\ &= (f(x=1, \boldsymbol{u}) \wedge g(x=1, \boldsymbol{u})) \oplus (f(x=0, \boldsymbol{u}) \wedge g(x=0, \boldsymbol{u})) \end{aligned}$$

Because

$$A \oplus B = \neg A \wedge B \vee \neg B \wedge A$$

, So

$$\begin{aligned} D_x(f \wedge g) = (&\neg f(x=1, \boldsymbol{u}) \vee \neg g(x=1, \boldsymbol{u}) \wedge f(x=0, \boldsymbol{u}) \wedge g(x=0, \boldsymbol{u}) \\ &\vee (\neg f(x=0, \boldsymbol{u}) \vee \neg g(x=0, \boldsymbol{u}) \wedge f(x=1, \boldsymbol{u}) \wedge g(x=1, \boldsymbol{u}) \end{aligned}$$

2 问题 2：XXX

附录：AI 使用记录